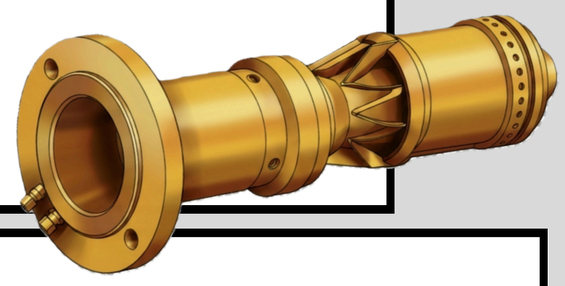
Accelerating Premixed H_2–Air Combustion Simulations Using Neural ODE

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INTRODUCTION :

- Hydrogen is gaining attention as a clean fuel as it produces no CO_2/greenhouse gases during combustion and has high energy conversion potential.
- Laminar flame speed (LFS), defined as the propagation speed of a planar, unstretched premixed flame, is a fundamental parameter for validating any chemical reaction mechanism.
- Initially, hydrogen–air combustion was simulated in ANSYS Fluent to determine LFS.
- Thereafter, to reduce the high computational cost of detailed chemistry, a Neural Ordinary Differential Equation (NODE) model was integrated into the workflow, providing faster source-term predictions while maintaining good accuracy.

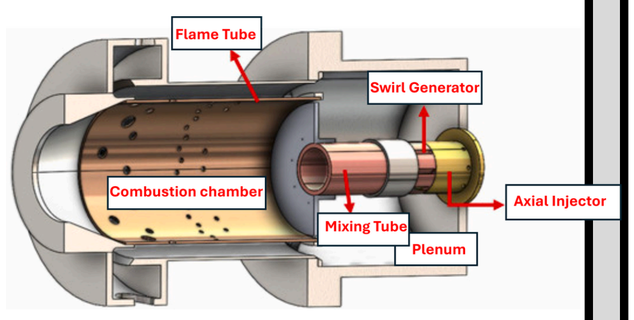
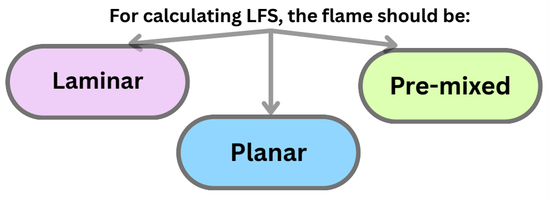
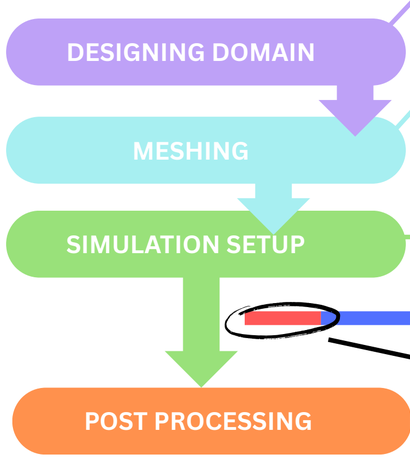


Fig: Representative schematic of a hydrogen–air combustor used in practical combustion systems.[1]



METHEDOLOGY :



- A 2D computational domain of dimensions 50 mm × 0.1 mm was created using ANSYS DesignModeler[2].
- The domain was divided into 3000 longitudinal sections, and 5 equal transverse divisions.

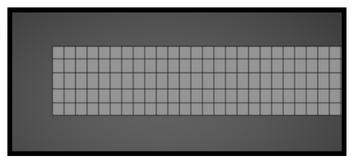


Fig: Zoomed view of Mesh

- Domain was initialized with a premixed H_2–air mixture at 300 K for different equivalence ratios.
- Combustion initiated by a 2 mm, 1800K patch near the left boundary
- Laminar flame speed was calculated from the Temperature vs. position profiles using:

$$S_L = \frac{x_2 - x_1}{t_2 - t_1}$$

RESULTS :

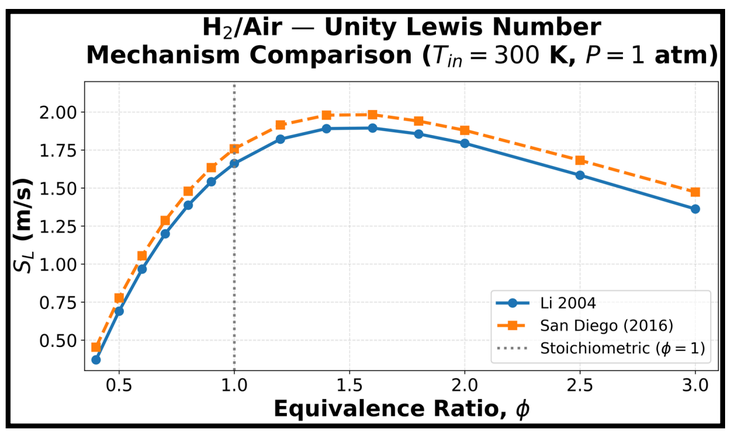


Fig: Comparison of laminar flame speed variation with Φ between the Li (2004) and San Diego (2016) reaction mechanisms (unity Lewis number, 300 K, 1 atm)

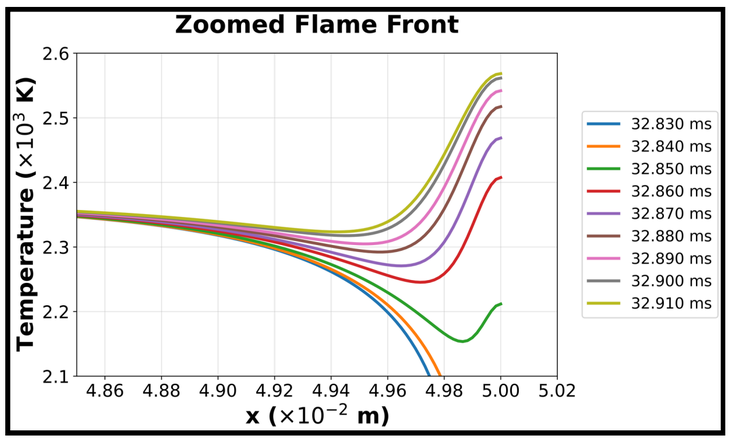
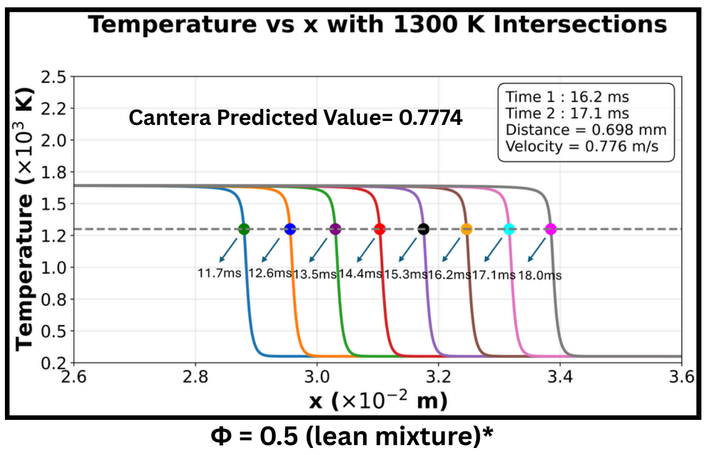
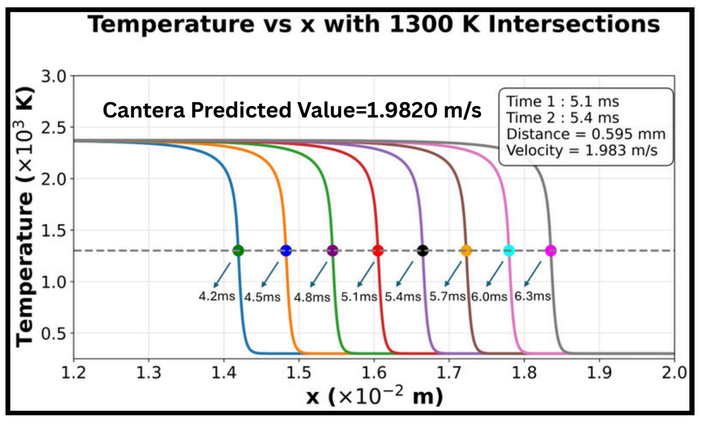


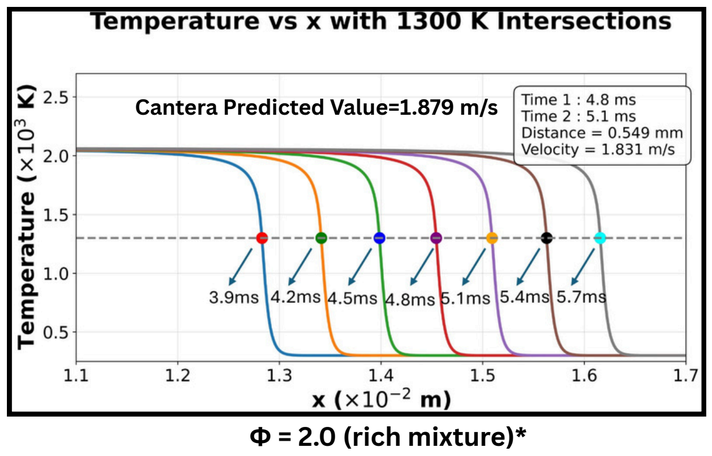
Fig: Flames chrashing after hitting the right wall of combustion chamber for Φ=1.6



Φ = 0.5 (lean mixture)*



Φ = 1.6 (rich mixture)*



Φ = 2.0 (rich mixture)*

*Simulations performed using the San Diego chemical kinetic mechanism (9 species, 21 elementary reactions).

IMPLEMENTATION OF NEURAL ODE:

A Neural Ordinary Differential Equation (Neural-ODE) coupled with a Detailed Kinetic Mechanism (DKM) was used to replace the computationally expensive finite-rate chemistry solver.[4]

9 species+Temperature are collected at every cell of the domain

$$x_n = [Y_1, Y_2, \ldots, Y_9, T_n]^T$$

The temperature is normalized using:

$$T_{\text{norm}} = \frac{T - T_{\min}}{T_{\max} - T_{\min}}$$

And, the species mass-fractions are normalized as:

$$Y_{\log} = \ln(\max(Y, \varepsilon)) \qquad Y_{\text{norm}} = \frac{Y_{\log} - \mu_Y}{\sigma_Y}$$

These values are then compressed to 5D latent space and passed to the NODE, latent state was advanced over one chemical time step by solving:

$$\frac{dz}{dt} = f_\theta(z)$$

The decoder then reconstructed the thermochemical state :

$$x_{n+1} = [Y_{1,n+1}, Y_{2,n+1}, \ldots, Y_{9,n+1}, T_{n+1}]^T$$

And, the temperature was reconstructed using mixture enthalpy conservation:

$$h(T_n, Y_n) = h(T_{n+1}, Y_{n+1})$$

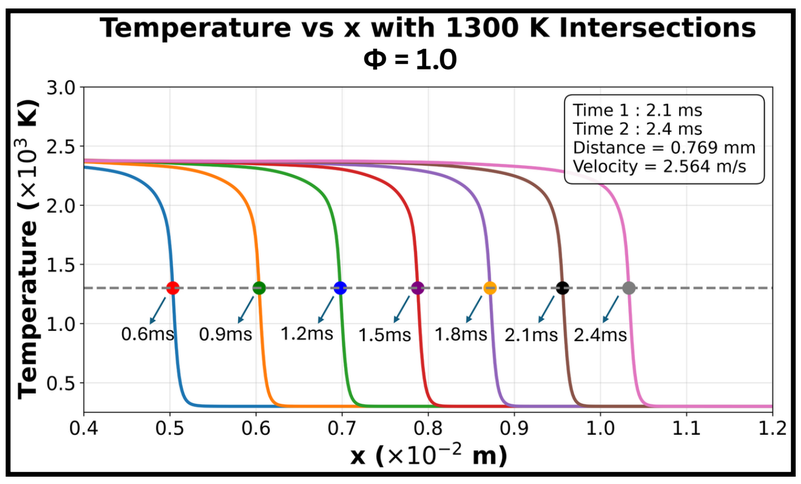
The reconstructed state was used to compute the source terms:

$$\omega_i = \rho \frac{Y_{i,n+1} - Y_{i,n}}{\Delta t}$$

The heat transfer rate is calculated using this formula:

$$\dot{Q} = -\sum_i \omega_i h_{f,i}$$

Mechanism Used:
San Diego mechanism (9 species, 21 elementary reactions)
9 species are: H_2, O_2, H, O, OH, H_2O, HO_2, H_2O_2, and N_2



Temperature v/s position profiles obtained from the Neural ODE–ANSYS Fluent coupled framework at Φ = 1.0. The flame is still in the transient stage and has not yet reached its steady laminar flame speed; simulations are currently in progress.

CONCLUSION:

- Laminar flame speed increased from the lean mixture (Φ = 0.5) to a maximum at Φ = 1.6 and then decreased for Φ = 2.0, which is consistent with established H_2–air combustion behavior and matches with the results obtained from Cantera.
- The maximum adiabatic flame temperature was obtained at Φ = 1.6, coinciding with the highest laminar flame speed.
- States with reconstruction errors above 2% were automatically solved using DKM, ensuring robust and reliable predictions for each cell.
- The coupled DKM–Neural ODE framework efficiently predicts chemical source terms while maintaining physical consistency in predicting temperature through enthalpy conservation.
- Neural-ODE predictions showed good agreement with Cantera/DKM results while significantly reducing the computational cost of combustion simulations.

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